

TIMBER BEAM N. 001

Caratteristiche meccaniche legno		
$id_{timber} = 1$		
$\gamma_M = 1.3 \quad k_{mod} = 0.9$		
$f_{m,k} = 20 \text{ MPa}$	$f_{m,d} = 13.85 \text{ MPa}$	$E_{0,mean} = 8400 \text{ MPa}$
$f_{t,0,k} = 16 \text{ MPa}$	$f_{t,0,d} = 11.08 \text{ MPa}$	$E_{0,05} = 7000 \text{ MPa}$
$f_{t,90,k} = 0.5 \text{ MPa}$	$f_{t,90,d} = 0.346 \text{ MPa}$	$E_{90,mean} = 300 \text{ MPa}$
$f_{c,0,k} = 20 \text{ MPa}$	$f_{c,0,d} = 13.85 \text{ MPa}$	$G_{mean} = 650 \text{ MPa}$
$f_{c,90,k} = 2.5 \text{ MPa}$	$f_{c,90,d} = 1.73 \text{ MPa}$	$r_k = 340 \text{ kg/m}^3$
$f_{v,k} = 3.5 \text{ MPa}$	$f_{v,d} = 2.42 \text{ MPa}$	$r_{mean} = 370 \text{ kg/m}^3$

Caratteristiche sezione rettangolare	
$b = 120 \text{ mm} \quad h = 240 \text{ mm} \quad \gamma = 5 \frac{\text{kN}}{\text{m}^3}$	
$A = 288 \text{ cm}^2$	$g_{1,k} = 0.144 \text{ kN/m}$
$I_y = 13824 \text{ cm}^4$	$I_z = 3456 \text{ cm}^4$
$W_y = 1152 \text{ cm}^3$	$W_z = 576 \text{ cm}^3$
$i_y = 6.93 \text{ cm}$	$i_z = 3.46 \text{ cm}$

$l = 8.79 \text{ m}$
 $k_m = 0.7$ legno massiccio, sezioni rettangolari
 $\beta_c = 0.1$ legno lammellare
 $l_{0,y} = 8.79 \text{ m}$
 $l_{0,z} = 8.79 \text{ m}$
 $\beta = 1$ coef. di vincolo

Azioni agenti

$M_{Ed,y} = 218.47 \text{ kNm}$
 $M_{Ed,z} = 0 \text{ kNm}$
 $N_{Ed} = 0 \text{ kN}$
 $V_{Ed,y} = 0 \text{ kN}$
 $V_{Ed,z} = 73.27 \text{ kN}$

Tensioni agenti

$\sigma_{c,0,d} = \frac{N_{Ed}}{A} = \frac{0 \text{ kN}}{288 \text{ cm}^2} = 0 \text{ MPa}$
 $\sigma_{m,y,d} = \frac{M_{Ed,y}}{W_y} = \frac{218.47 \text{ kNm}}{1152 \text{ cm}^3} = 189.64 \text{ MPa}$
 $\sigma_{m,z,d} = \frac{M_{Ed,z}}{W_z} = \frac{0 \text{ kNm}}{576 \text{ cm}^3} = 0 \text{ MPa}$

$$k_{cr} = \frac{2.5 \text{ MPa}}{f_{v,k}} = \frac{2.5 \text{ MPa}}{3.5 \text{ MPa}} = 0.714$$

$$\tau_{y,d} = \frac{1.5 \cdot V_{Ed,y}}{k_{cr} \cdot A} = \frac{1.5 \cdot 0 \text{ kN}}{0.714 \cdot 288 \text{ cm}^2} = 0 \text{ MPa}$$

$$\tau_{z,d} = \frac{1.5 \cdot V_{Ed,z}}{k_{cr} \cdot A} = \frac{1.5 \cdot 73.27 \text{ kN}}{0.714 \cdot 288 \text{ cm}^2} = 5.34 \text{ MPa}$$

Verifica flessione

$$\frac{\sigma_{m,y,d}}{f_{m,d}} + k_m \cdot \frac{\sigma_{m,z,d}}{f_{m,d}} = \frac{189.64 \text{ MPa}}{13.85 \text{ MPa}} + 0.7 \cdot \frac{0 \text{ MPa}}{13.85 \text{ MPa}} = 1369.65 \% \text{ NO } \times$$

$$k_m \cdot \frac{\sigma_{m,y,d}}{f_{m,d}} + \frac{\sigma_{m,z,d}}{f_{m,d}} = 0.7 \cdot \frac{189.64 \text{ MPa}}{13.85 \text{ MPa}} + \frac{0 \text{ MPa}}{13.85 \text{ MPa}} = 958.76 \% \text{ NO } \times$$

Verifica flessione e compressione combinata

$$\left(\frac{\sigma_{c,0,d}}{f_{c,0,d}} \right)^2 + \frac{\sigma_{m,y,d}}{f_{m,d}} + k_m \cdot \frac{\sigma_{m,z,d}}{f_{m,d}} = \left(\frac{0 \text{ MPa}}{13.85 \text{ MPa}} \right)^2 + \frac{189.64 \text{ MPa}}{13.85 \text{ MPa}} + 0.7 \cdot \frac{0 \text{ MPa}}{13.85 \text{ MPa}} = 1369.65 \% \text{ NO } \times \quad [\text{EC5 §6.2.4}]$$

$$\left(\frac{\sigma_{c,0,d}}{f_{c,0,d}} \right)^2 + k_m \cdot \frac{\sigma_{m,y,d}}{f_{m,d}} + \frac{\sigma_{m,z,d}}{f_{m,d}} = \left(\frac{0 \text{ MPa}}{13.85 \text{ MPa}} \right)^2 + 0.7 \cdot \frac{189.64 \text{ MPa}}{13.85 \text{ MPa}} + \frac{0 \text{ MPa}}{13.85 \text{ MPa}} = 958.76 \% \text{ NO } \times$$

Verifica a taglio

$$\frac{\tau_{y,d}}{f_{v,d}} = \frac{0 \text{ MPa}}{2.42 \text{ MPa}} = 0 \% \text{ Ok } \checkmark$$

$$\frac{\tau_{z,d}}{f_{v,d}} = \frac{5.34 \text{ MPa}}{2.42 \text{ MPa}} = 220.49 \% \text{ NO } \times$$

Verifica instabilità di colonna

$$\lambda_y = \frac{l_{0,y}}{i_y} = \frac{8.79 \text{ m}}{6.93 \text{ cm}} = 126.81 \quad [\text{EC5 §6.3.2}]$$

$$\lambda_{rel,y} = \frac{\lambda_y}{\pi} \cdot \sqrt{\frac{f_{c,0,k}}{E_{0,05}}} = \frac{126.81}{3.14} \cdot \sqrt{\frac{20 \text{ MPa}}{7000 \text{ MPa}}} = 2.16$$

$$k_y = 0.5 \cdot (1 + \beta_c \cdot (\lambda_{rel,y} - 0.3) + \lambda_{rel,y}^2) = 0.5 \cdot (1 + 0.1 \cdot (2.16 - 0.3) + 2.16^2) = 2.92$$

$$k_{c,y} = \frac{1}{k_y + \sqrt{k_y^2 - \lambda_{rel,y}^2}} = \frac{1}{2.92 + \sqrt{2.92^2 - 2.16^2}} = 0.205$$

$$\lambda_z = \frac{l_{0,z}}{i_z} = \frac{8.79 \text{ m}}{3.46 \text{ cm}} = 253.63$$

$$\lambda_{rel,z} = \frac{\lambda_z}{\pi} \cdot \sqrt{\frac{f_{c,0,k}}{E_{0,05}}} = \frac{253.63}{3.14} \cdot \sqrt{\frac{20 \text{ MPa}}{7000 \text{ MPa}}} = 4.32$$

$$k_z = 0.5 \cdot (1 + \beta_c \cdot (\lambda_{rel,z} - 0.3) + \lambda_{rel,z}^2) = 0.5 \cdot (1 + 0.1 \cdot (4.32 - 0.3) + 4.32^2) = 10.01$$

$$k_{c,z} = \frac{1}{k_z + \sqrt{k_z^2 - \lambda_{rel,z}^2}} = \frac{1}{10.01 + \sqrt{10.01^2 - 4.32^2}} = 0.0525$$

$$\frac{\sigma_{c,0,d}}{k_{c,y} \cdot f_{c,0,d}} + \frac{\sigma_{m,y,d}}{f_{m,d}} + k_m \cdot \frac{\sigma_{m,z,d}}{f_{m,d}} = \frac{0 \text{ MPa}}{0.205 \cdot 13.85 \text{ MPa}} + \frac{189.64 \text{ MPa}}{13.85 \text{ MPa}} + 0.7 \cdot \frac{0 \text{ MPa}}{13.85 \text{ MPa}} = 1369.65 \% \text{ NO } \times$$

$$\frac{\sigma_{c,0,d}}{k_{c,z} \cdot f_{c,0,d}} + k_m \cdot \frac{\sigma_{m,y,d}}{f_{m,d}} + \frac{\sigma_{m,z,d}}{f_{m,d}} = \frac{0 \text{ MPa}}{0.0525 \cdot 13.85 \text{ MPa}} + 0.7 \cdot \frac{189.64 \text{ MPa}}{13.85 \text{ MPa}} + \frac{0 \text{ MPa}}{13.85 \text{ MPa}} = 958.76 \% \text{ NO } \times$$

Verifica instabilità di trave

$$l_{ef} = \beta \cdot l = 1 \cdot 8.79 \text{ m} = 8.79 \text{ m} \quad [\text{EC5 §6.3.3}]$$

$$\sigma_{m,crit} = \frac{0.78 \cdot b^2}{h \cdot l_{ef}} \cdot E_{0,05} = \frac{0.78 \cdot (120 \text{ mm})^2}{240 \text{ mm} \cdot 8.79 \text{ m}} \cdot 7000 \text{ MPa} = 37.29 \text{ MPa}$$

$$\lambda_{rel,m} = \sqrt{\frac{f_{m,k}}{\sigma_{m,crit}}} = \sqrt{\frac{20 \text{ MPa}}{37.29 \text{ MPa}}} = 0.732$$

$$k_{crit} = \begin{cases} \text{if } \lambda_{rel,m} < 0.75: 1 \\ \text{else if } \lambda_{rel,m} < 1.4: 1.56 - 0.75 \cdot \lambda_{rel,m} \\ \text{else: } \frac{1}{\lambda_{rel,m}^2} \end{cases} = \begin{cases} \text{if } 0.732 < 0.75: 1 \\ \text{else if } 0.732 < 1.4: 1.56 - 0.75 \cdot 0.732 \\ \text{else: } \frac{1}{0.732^2} \end{cases} = 1$$

$$\frac{\sigma_{m,y,d}}{k_{crit} \cdot f_{m,d}} = \frac{189.64 \text{ MPa}}{1 \cdot 13.85 \text{ MPa}} = 1369.65 \% \text{ NO } \times$$

$$\left(\frac{\sigma_{m,y,d}}{k_{crit} \cdot f_{m,d}} \right)^2 + \frac{\sigma_{c,0,d}}{k_{c,z} \cdot f_{c,0,d}} = \left(\frac{189.64 \text{ MPa}}{1 \cdot 13.85 \text{ MPa}} \right)^2 + \frac{0 \text{ MPa}}{0.0525 \cdot 13.85 \text{ MPa}} = 18759.5 \% \text{ NO } \times$$

Freccia

$$q_G = 8.64 \frac{\text{kN}}{\text{m}}$$

$$q_{Q1} = 10.92 \frac{\text{kN}}{\text{m}}$$

$$k_{def} = 0.6$$

$$\psi_{21} = 0.2$$

$$w_{lim,inst} = \frac{l}{300} = \frac{8.79 \text{ m}}{300} = 29.29 \text{ mm}$$

$$w_{lim,fin} = \frac{l}{200} = \frac{8.79 \text{ m}}{200} = 43.93 \text{ mm}$$

$$w(q) = \frac{5}{384} \cdot \frac{q \cdot l^4}{E_{0,mean} \cdot I_y} + 1.2 \cdot \frac{1}{8} \cdot \frac{q \cdot l^2}{G_{mean} \cdot A}$$

$$w \left(1 \frac{\text{kN}}{\text{m}} \right) = 67.44 \text{ mm}$$

$$w_{inst,G} = w(q_G) = w(8.64 \text{ kN/m}) = 582.65 \text{ mm}$$

$$w_{inst,Q1} = w(q_{Q1}) = w(10.92 \text{ kN/m}) = 736.4 \text{ mm}$$

$$w_{inst,TOT} = w_{inst,G} + w_{inst,Q1} = 582.65 \text{ mm} + 736.4 \text{ mm} = 1319.05 \text{ mm}$$

$$w_{inst,TOT} = 1319.05 \text{ mm} > w_{lim,inst} = 29.29 \text{ mm} \text{ NO } \times$$

$$w_{fin,G1} = w_{inst,G} \cdot (1 + k_{def}) = 582.65 \text{ mm} \cdot (1 + 0.6) = 932.24 \text{ mm}$$

$$w_{fin,Q1} = w_{inst,Q1} \cdot (1 + \psi_{21} \cdot k_{def}) = 736.4 \text{ mm} \cdot (1 + 0.2 \cdot 0.6) = 824.77 \text{ mm}$$

$$w_{fin,TOT} = w_{fin,G1} + w_{fin,Q1} = 932.24 \text{ mm} + 824.77 \text{ mm} = 1757.01 \text{ mm}$$

$$w_{fin,TOT} = 1757.01 \text{ mm} > w_{lim,fin} = 43.93 \text{ mm} \text{ NO } \times$$